PORTFOLIO ANALYSIS

Portfolio Performance and Diversification

VBTLX + VFIAX compared with AAPL

January–July 2016 | Hypothetical $5 million investment

Client takeaway

AAPL delivered more return in this short window; the mixed portfolio delivered a much smoother ride.

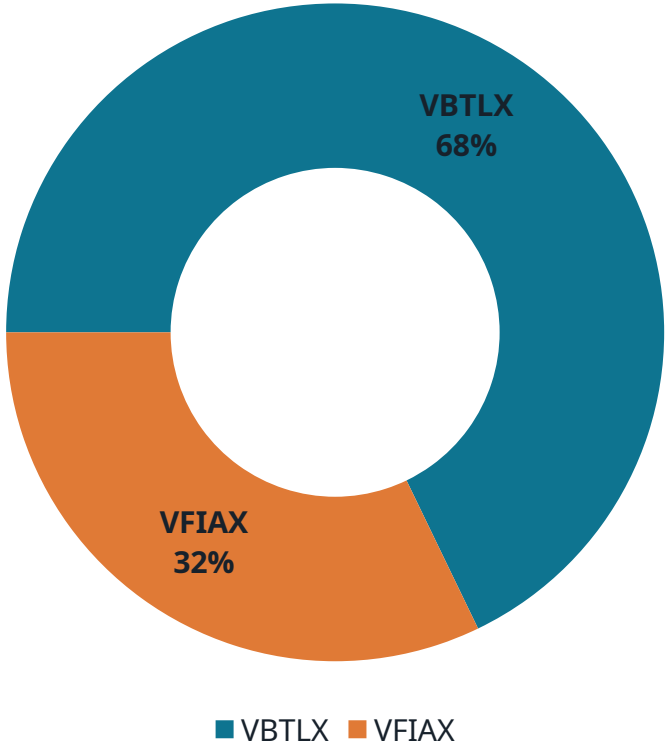
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Business and Financial Modeling Capstone

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A bond-led allocation maximizes the historical Sharpe ratio

Estimation period: January 2012–December 2015; risk-free rate assumed to be 0%.



VBTLX · TOTAL BOND MARKET

67.85%

$3,392,295 of the initial $5m

VFIAX · S&P 500 INDEX

32.15%

$1,607,705 of the initial $5m

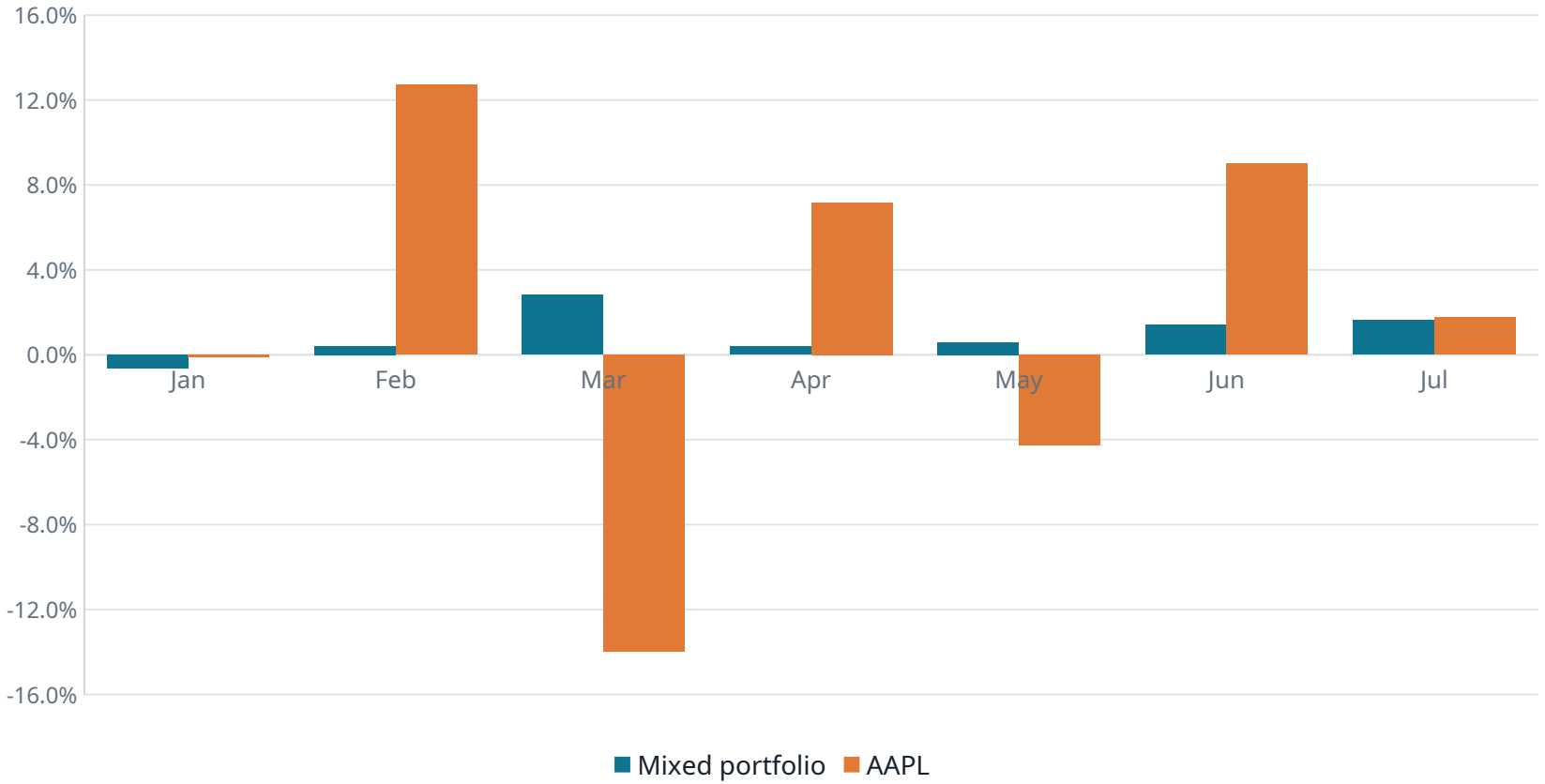
Why this mix?

The allocation balances equity upside with bond-market stability and diversification.

Source: course-provided monthly return files; calculations use adjusted returns.

Diversification sharply reduced month-to-month swings

Monthly total return, January–July 2016.



AAPL RANGE

26.7 pts

+12.7% to −14.0%

MIXED RANGE

3.4 pts

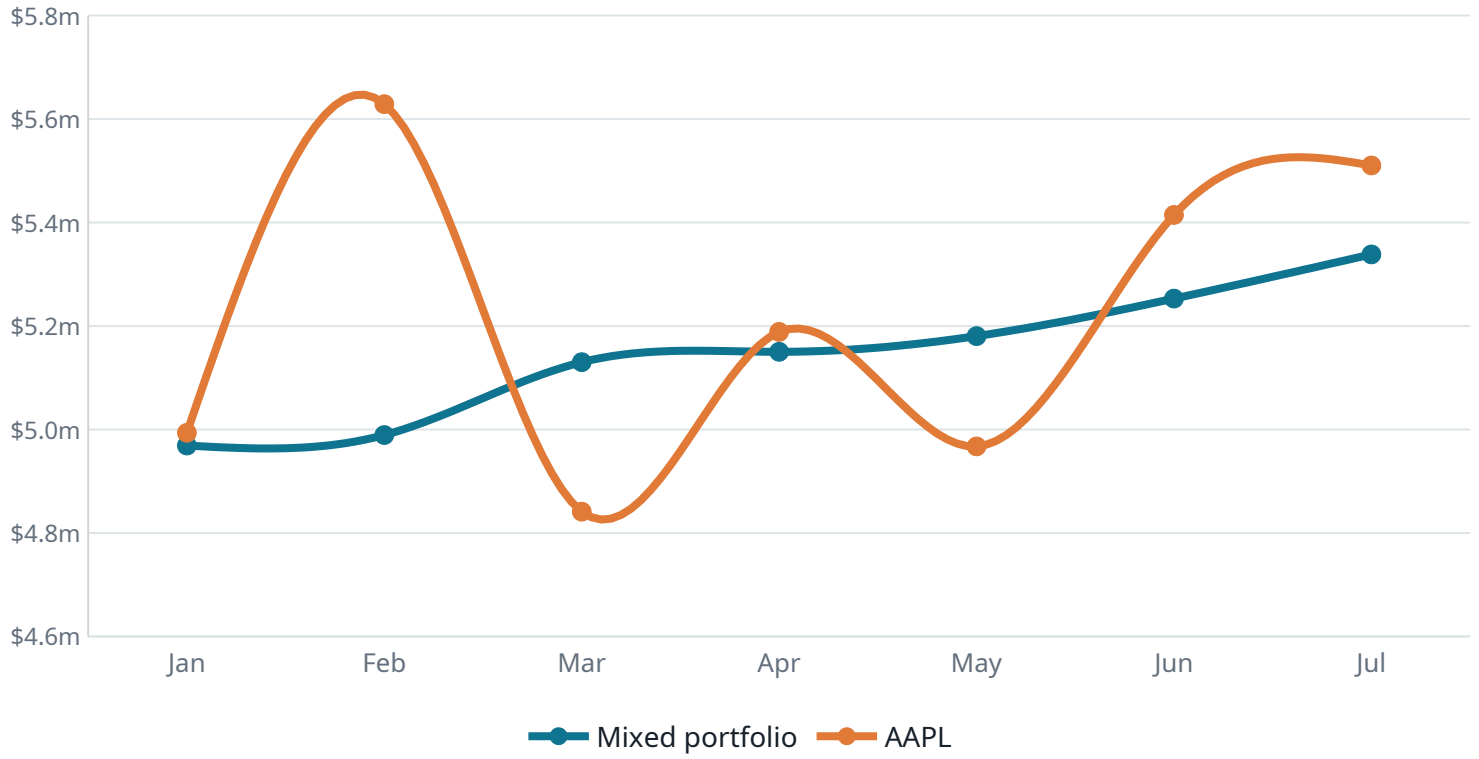
+2.8% to −0.6%

The mixed portfolio was positive in six of seven months.

Source: course-provided monthly return files; calculations use adjusted returns.

AAPL finished higher, but the path was substantially riskier

Growth of a hypothetical $5 million investment; monthly rebalancing weights held constant.



CUMULATIVE RETURN

+6.77% vs +10.21%

Mixed portfolio vs AAPL

MONTHLY VOLATILITY

1.11% vs 9.04%

AAPL was 8.15× more volatile

WORST MONTH

−0.62% vs −13.99%

Mixed portfolio vs AAPL

Source: course-provided monthly return files; calculations use adjusted returns.

Choose the allocation that matches the client's capacity for loss

What the evidence shows

1 AAPL produced the higher seven-month return.

2 The mixed portfolio cut monthly volatility by about 88%.

3 Its worst month was limited to −0.62%, versus −13.99% for AAPL.

Recommended action

Use the diversified VBTLX/VFIAX mix for clients who prioritize capital stability and smoother outcomes. Increase VFIAX only when risk tolerance and investment horizon support it.

Monitor correlations, rebalance periodically, and stress-test the mix before changing strategic weights.

Important: this seven-month comparison is historical and does not guarantee future performance.